BUSINESS ANALYTICS · TASK 2

Enterprise Sales Performance & Financial Analysis

Power Query ETL, star-schema data model with DAX, financial model, forecast, what-if analysis and an interactive dashboard — India + UAE, 2024–2025

₹30,921,829	+14%	4 → 1	15
FY2025 revenue	YoY growth	source layouts unified	DAX measures

ANALYST
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PROGRAMME
Sumerix Global · Business Analytics · 36 weeks

BATCH START
15 October 2026

TOOLS
Excel (Tables, SUMIFS, XLOOKUP, Power Query, Power Pivot / DAX)

1 · Executive summary

₹30,921,829
FY2025 net revenue (+14% YoY)

28.7%
gross margin

₹2,236,196
net profit (7.2% margin)

₹33,465,339
FY2026 forecast

FY2025 net revenue ₹30,921,829 (+14% YoY), gross margin 28.7%, net profit ₹2,236,196 (7.2% net margin).

Industry challenge. Four branch extracts arrive in four layouts (CSV, Excel, semicolon CSV, AED prices) with text dates, duplicates and blanks; manual consolidation took days. Power Query now refreshes all of it in one click.

Dataset. DECLARED SYNTHETIC — 6,839 order lines (Jan 2024 – Dec 2025) for a fictional retailer with 8 stores in India and 2 in the UAE, generated with a fixed seed. AED is converted to INR at assumed annual average rates (2024: 22.70, 2025: 23.45).

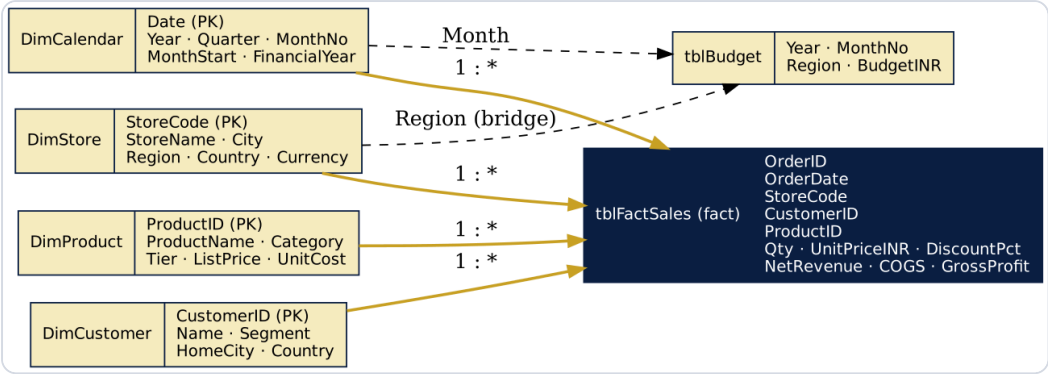
2 · Power Query ETL — four layouts, one refresh

Query	Source	Applied steps	Rows	Load to
`qry_Sales_North`	sales_india_north.csv (CSV, ISO dates, INR)	Promote headers ▸ Change types ▸ Rename UnitPrice→UnitPriceINR ▸ Add SourceFile	1756	Connection only
`qry_Sales_South`	sales_india_south.xlsx (Excel sheet 'Sales', dd-mm-yyyy text dates, Disc % as whole numbers)	Rename 9 columns ▸ Date.FromText(dd-MM-yyyy) ▸ Disc % ÷ 100 ▸ Add SourceFile	2053	Connection only
`qry_Sales_West`	sales_india_west.csv (semicolon-delimited, dd/mm/yyyy, UPPERCASE channel, duplicates, blanks)	Split by ';' ▸ Rename ▸ Date (en-GB) ▸ Text.Proper(channel) ▸ Replace '' → 'Unknown' ▸ Remove Duplicates	1420	Connection only
`qry_Sales_UAE`	sales_uae.csv (prices in AED)	Rename ▸ Merge fx_rates on Year ▸ UnitPriceINR = PriceAED × Rate ▸ Remove helper columns	1610	Connection only
`qry_Sales_All`	Append of the four branch queries	Table.Combine ▸ Merge Dim_Store (Region, Country) ▸ Merge Dim_Product (Category, UnitCostINR) ▸ Merge Dim_Customer (Segment) ▸ Sort	6839	Table + Data Model
`qry_Customers` / `qry_Products` / `qry_Stores`	reference CSVs	Promote headers ▸ Change types ▸ Remove errors ▸ Trim text	600 / 16 / 10	Table + Data Model
`qry_Budget`	budget_2025.csv	Change types ▸ Unpivot not required (already long format)	48	Table + Data Model

`qry_Calendar`	generated with List.Dates	List.Dates(#date(2024,1,1), 731, #duration(1,0,0,0)) ▸ Add Year/Quarter/Month/FY columns	731	Table + Data Model

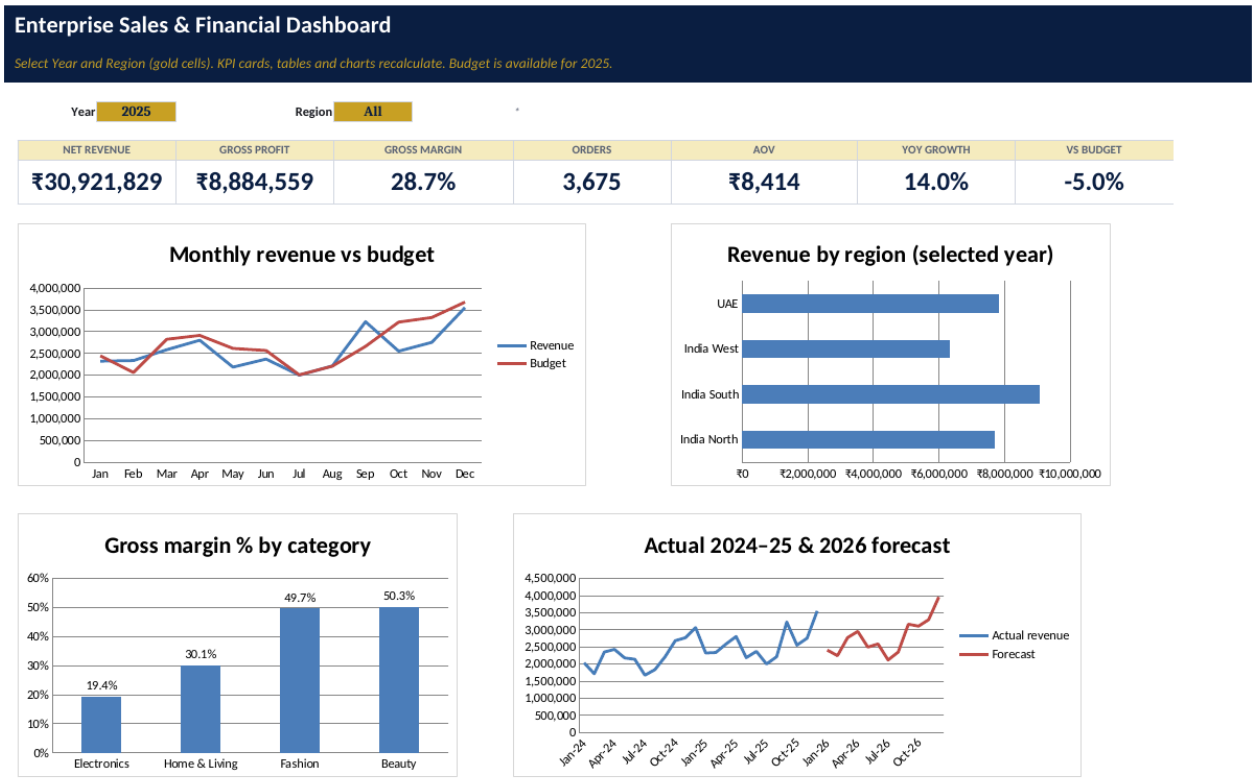
All M code is in `power-query/*.pq`. Data-quality checks after refresh: 6,839 rows = 6,839 distinct OrderIDs, 0 lines without a region after the merge, 3 blank West channels mapped to *Unknown*, 1,610 UAE lines converted from AED.

3 · Data model — star schema & DAX



Measure	DAX
Total Revenue	`SUM ( tblFactSales[NetRevenue] )`
Total COGS	`SUM ( tblFactSales[COGS] )`
Gross Profit	`[Total Revenue] - [Total COGS]`
Gross Margin %	`DIVIDE ( [Gross Profit], [Total Revenue] )`
Orders	`DISTINCTCOUNT ( tblFactSales[OrderID] )`
Average Order Value	`DIVIDE ( [Total Revenue], [Orders] )`
Active Customers	`DISTINCTCOUNT ( tblFactSales[CustomerID] )`
Revenue LY	`CALCULATE ( [Total Revenue], SAMEPERIODLASTYEAR ( DimCalendar[Date] ) )`
Revenue YoY %	`DIVIDE ( [Total Revenue] - [Revenue LY], [Revenue LY] )`
Revenue YTD	`TOTALYTD ( [Total Revenue], DimCalendar[Date] )`
Budget	`CALCULATE ( SUM ( tblBudget[BudgetINR] ), TREATAS ( SUMMARIZE ( DimCalendar, DimCalendar[Year], DimCalendar[MonthNo] ), tblBudget[Year], tblBudget[MonthNo] ), TREATAS ( VALUES ( DimStore[Region] ), tblBudget[Region] ) )`
Budget Variance	`[Total Revenue] - [Budget]`
Budget Variance %	`DIVIDE ( [Budget Variance], [Budget] )`
Fixed Opex	`SUMX ( VALUES ( DimCalendar[MonthStart] ), SWITCH ( YEAR ( DimCalendar[MonthStart] ), 2024, 240000, 2025, 260000, 0 ) )`
Net Profit (pre-tax)	`[Gross Profit] - [Total Revenue] * 0.09 - [Fixed Opex]`

4 · Interactive dashboard



Year = 2025, Region = All. KPI cards: net revenue, gross profit, gross margin, orders, AOV, YoY growth and variance to budget.

Regional performance 2025

Region	Revenue 2024	Revenue 2025	YoY	Budget 2025	Attainment	GP %	Rank	Status
India North	₹6,392,659	₹7,713,417	20.7%	₹7,669,000	100.6%	29.1%	3	On / above budget
India South	₹8,707,058	₹9,060,205	4.1%	₹10,449,000	86.7%	29.1%	1	Below budget
India West	₹5,920,965	₹6,333,405	7.0%	₹7,106,000	89.1%	28.6%	4	Below budget
UAE	₹6,098,388	₹7,814,803	28.1%	₹7,319,000	106.8%	28.0%	2	On / above budget

5 · Category & store analysis

Category	Rev 2025	YoY	GP %	Share
Electronics	₹17,144,578	11.8%	19.4%	55.4%
Home & Living	₹6,623,004	25.2%	30.1%	21.4%
Fashion	₹4,557,802	6.5%	49.7%	14.7%
Beauty	₹2,596,445	17.0%	50.3%	8.4%

Store	Name	H1-2025	H2-2025	H2 vs H1
N01	Connaught Place	₹1,513,583	₹1,575,619	4.1%
N02	Raj Nagar	₹1,012,748	₹1,205,895	19.1%
N03	Sector 18	₹1,174,343	₹1,231,228	4.8%
S01	T. Nagar	₹1,617,811	₹2,012,890	24.4%
S02	RS Puram	₹918,659	₹1,212,392	32.0%
S03	Indiranagar	₹1,552,263	₹1,746,190	12.5%
W01	Bandra West	₹1,846,157	₹2,123,561	15.0%
W02	Koregaon Park	₹1,070,776	₹1,292,911	20.7%
U01	Al Karama	₹2,152,740	₹2,114,737	-1.8%
U02	Al Nahda	₹1,755,830	₹1,791,496	2.0%

6 · Financial model

P&L (₹)	FY2024 actual	FY2025 actual	FY2026 plan	YoY 25 vs 24	YoY 26 vs 25
Net revenue	₹27,119,070	₹30,921,829	₹35,034,433	14.0%	13.3%
Cost of goods sold	₹19,472,720	₹22,037,270	₹25,210,637	13.2%	14.4%
Gross profit	₹7,646,350	₹8,884,559	₹9,823,796	16.2%	10.6%
Gross margin %	28.2%	28.7%	28.0%		
Variable opex	₹2,440,716	₹2,782,965	₹3,153,099	14.0%	13.3%
Fixed opex	₹2,880,000	₹3,120,000	₹3,369,600	8.3%	8.0%
Operating profit (EBIT)	₹2,325,633	₹2,981,595	₹3,301,097	28.2%	10.7%
Tax	₹581,408	₹745,399	₹825,274	28.2%	10.7%
Net profit	₹1,744,225	₹2,236,196	₹2,475,823	28.2%	10.7%
Net margin %	6.4%	7.2%	7.1%		

Break-even (FY2026 plan)

Metric	Value
Contribution margin ratio	19.0%
Fixed costs (annual)	₹3,369,600
Break-even revenue	₹17,697,106
Margin of safety	49.5%

Investment appraisal — ₹25L WMS + sortation

Metric	Value
NPV @ 12%	₹260,749
IRR	16.0%
Payback	3.4 years
Loan EMI (60 m @ 10.5%)	₹53,735

Savings assumption: 2.0% of FY2026 plan revenue, growing 5% a year (declared input).

7 · Forecast & what-if analysis



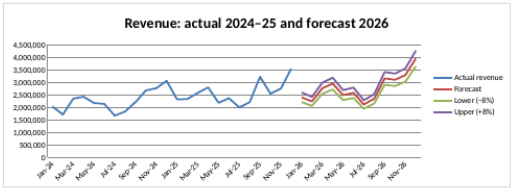
Forecast — 2026 monthly revenue (trend × seasonality) + Excel Forecast Sheet steps

Seasonal index from 2 years of actuals; trend from FORECAST.LINEAR on de-seasonalised revenue

Period #	Month	Actual revenue	Seasonal index	De-seasonalised	Trend (FORECAST.LINEAR)	Forecast	Lower (−8%)	Upper (+8%)	si ratio	Month	Year
1	Jan-24	₹2,037,640	0.902	₹2,259,408	₹2,186,530				0.902	1	2024
2	Feb-24	₹1,720,510	0.835	₹2,061,579	₹2,206,712				0.781	2	2024
3	Mar-24	₹2,353,834	1.023	₹2,301,815	₹2,226,893				1.042	3	2024
4	Apr-24	₹2,429,918	1.082	₹2,246,166	₹2,247,074				1.075	4	2024
5	May-24	₹2,179,804	0.907	₹2,403,766	₹2,267,256				0.965	5	2024
6	Jun-24	₹2,141,691	0.934	₹2,292,580	₹2,287,437				0.948	6	2024
7	Jul-24	₹1,675,590	0.759	₹2,206,436	₹2,307,619				0.741	7	2024
8	Aug-24	₹1,841,039	0.837	₹2,199,613	₹2,327,800				0.815	8	2024
9	Sep-24	₹2,218,495	1.117	₹1,985,527	₹2,347,982				0.982	9	2024
10	Oct-24	₹2,683,856	1.089	₹2,463,929	₹2,368,163				1.188	10	2024
11	Nov-24	₹2,770,498	1.148	₹2,413,980	₹2,388,345				1.226	11	2024
12	Dec-24	₹3,066,194	1.368	₹2,242,189	₹2,408,526				1.357	12	2024
13	Jan-25	₹2,324,426	0.902	₹2,577,406	₹2,428,707				0.902	1	2025
14	Feb-25	₹2,339,250	0.835	₹2,802,976	₹2,448,889				0.908	2	2025
15	Mar-25	₹2,586,205	1.023	₹2,529,052	₹2,469,070				1.004	3	2025
16	Apr-25	₹2,804,590	1.082	₹2,592,504	₹2,489,252				1.088	4	2025
17	May-25	₹2,188,002	0.907	₹2,412,806	₹2,509,433				0.849	5	2025
18	Jun-25	₹2,372,437	0.934	₹2,539,582	₹2,529,615				0.921	6	2025
19	Jul-25	₹2,003,176	0.759	₹2,637,805	₹2,549,796				0.777	7	2025
20	Aug-25	₹2,214,310	0.837	₹2,645,585	₹2,569,978				0.859	8	2025
21	Sep-25	₹3,228,749	1.117	₹2,889,692	₹2,590,159				1.253	9	2025
22	Oct-25	₹2,553,447	1.089	₹2,344,206	₹2,610,340				0.991	10	2025
23	Nov-25	₹2,755,784	1.148	₹2,401,159	₹2,630,522				1.069	11	2025
24	Dec-25	₹3,551,453	1.368	₹2,597,040	₹2,650,703				1.378	12	2025
25	Jan-26		0.902		₹2,670,885	₹2,408,730	₹2,216,031	₹2,601,428			
26	Feb-26		0.835		₹2,691,066	₹2,245,855	₹2,066,186	₹2,425,523			
27	Mar-26		1.023		₹2,711,248	₹2,772,519	₹2,550,717	₹2,994,320			
28	Apr-26		1.082		₹2,731,429	₹2,954,879	₹2,718,489	₹3,191,270			
29	May-26		0.907		₹2,751,611	₹2,495,240	₹2,295,621	₹2,694,859			
30	Jun-26		0.934		₹2,771,792	₹2,589,364	₹2,382,214	₹2,796,513			
31	Jul-26		0.759		₹2,791,973	₹2,120,252	₹1,950,632	₹2,289,873			
32	Aug-26		0.837		₹2,812,155	₹2,353,726	₹2,165,428	₹2,542,024			
33	Sep-26		1.117		₹2,832,336	₹3,164,664	₹2,911,490	₹3,417,837			
34	Oct-26		1.089		₹2,852,518	₹3,107,130	₹2,858,560	₹3,355,701			
35	Nov-26		1.148		₹2,872,699	₹3,296,965	₹3,033,208	₹3,560,722			
36	Dec-26		1.368		₹2,892,881	₹3,956,016	₹3,639,535	₹4,272,497			

FY2026 forecast total	₹33,465,339
vs FY2025 actual	8.2%
vs FY2026 plan (Financi	-4.5%

Excel Forecast Sheet (ETS) alternative: select B5:C28 ▸ Data ▸ Forecast Sheet ▸ end 31-Dec-2026, seasonality 12, confidence 95% ▸ Create. Compare its total with C42.



Seasonal indices from 24 months of actuals × a FORECAST.LINEAR trend on de-seasonalised revenue give an FY2026 forecast of **₹33,465,339** (+8.2% vs FY2025), with a ±8% band.

Scenario	Revenue 2026	EBIT 2026	Net profit 2026	Net margin
Base	₹35,034,433	₹3,301,097	₹2,475,823	7.1%
Best	₹37,338,109	₹4,567,333	₹3,425,500	9.2%
Worst	₹31,540,266	₹1,442,946	₹1,082,209	3.4%

Goal Seek: the Base plan needs a **5.3%** price change to reach the ₹40L EBIT target (solved algebraically in the sheet and reproducible with Data ▸ What-If ▸ Goal Seek). One- and two-variable data tables show EBIT sensitivity to price and volume.

8 · Findings, recommendations & strategy

1. **Budget.** Company revenue landed at 95% of the 2025 stretch budget (2024 × 1.20). Weakest region: India South at 87%.
2. **Seasonality.** Oct–Nov festive months carry India; UAE peaks in Dec (Dubai Shopping Festival) and Mar–Apr (Ramadan/Eid) — budgets must be phased by region, not spread evenly.
3. **Mix.** Highest-margin category: Beauty (50%); lowest: Electronics (19%).
4. **Outlook.** Trend × seasonality forecast for FY2026 is ₹33,465,339 (+8% vs 2025); the Base plan needs 5.3% price change to hit the EBIT target of ₹4,000,000.

Recommendations

1. Fix the weakest Indian store early: Connaught Place (N01) grew only 4% in H2-2025 vs H1, while festive demand lifted the best Indian store by 32% — run a store P&L review and a local promotion before the next festive quarter.
2. Protect margin: keep Electronics discounting ≤10%; grow Fashion and Beauty attachment (highest GP%) through bundles and loyalty offers.
3. Approve the ₹25L automation project: NPV ₹260,749 at 12%, IRR 16%, payback 3.4 years — tie release of funds to a 3-month pilot at the busiest hub.

Future strategy. Move the model into Power BI (Task 3): reuse these Power Query steps and DAX measures, publish a monthly refresh, and add row-level security by region.

How to rebuild the live model in Excel

1. Create the `DataFolder` parameter and paste each `power-query/*.pq` script (README in that folder).
2. Load `qry_Sales_All`, dimensions, budget and calendar to the Data Model; mark `DimCalendar` as the date table.
3. Create the four 1:* relationships, then paste the measures from `dax/measures.dax`.
4. Data ▸ Refresh All when a new month's extracts land in `data/raw`.